Sahaj Bhadja

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Summary

Forward Deployed Engineer with 2+ years building integrations, **APIs**, internal tools, and production **LLM workflows** in **Python** and TypeScript, working directly with clients to translate workflows into solutions, troubleshoot issues, and ship deployments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Advanced NLP, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Built **ETL-style integrations** connecting FastAPI services, Supabase, SQL Server, and cloud storage for a deployed AI platform, using incremental loads and reconciliation checks to run 5K+ monthly jobs at 99%+ reliability.
- Designed multi-tenant storage on Azure SQL with **row-level security** for 500+ users, cutting dashboard latency from 1.2s to 350ms, and restructured async LLM orchestration processing 1K+ jobs at 3x throughput.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built an AI **document analysis** pipeline with secure PDF, DOCX, and ZIP ingestion, vision OCR, and LLM evaluation across GPT-4o, Claude, and Qwen-VL, processing 1K+ files per batch for 150+ instructors.
- Shipped React and Flask REST features end to end to production on AWS, handling 10K+ daily requests with stable peak-load responses and **debugging** OCR timeouts with fallback routing, cutting turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Troubleshot large-scale **API integrations** with Microsoft and GitHub stakeholders, tracing failures through logs and tests, then adding pagination, field filtering, and caching to cut response times 30%.
- Took requirements through deployment on cross-platform data-sharing microservices, building **workflow automation** and 20+ automated unit and integration tests that cut manual data entry 35% and technical debt 20% across releases.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** and Node.js microservices platform with JWT authentication, role-based access, and REST APIs for customer records, billing, and leads, syncing 2M+ records across locations in real time.
- Used Google Analytics and Azure SQL reporting on feature usage and drop-off points to guide **data-driven** fixes that cut resource wastage 42% and lifted lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Worked directly with the client to map 12+ integration points and **presented the end-to-end architecture** for approval, then built Flask booking, pricing, and coupon APIs with sub-200ms responses.
- Integrated card, wallet, and PayPal payment providers through **idempotent webhooks**, automating refunds and disputes to cut manual handling about 70%, and added wallet-signature authentication for passwordless login.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, Bash
Integrations & APIs: REST APIs, Webhooks, JWT, Role-Based Access, Node.js, Express.js, FastAPI, Flask, React, Jest
AI & Automation: LLM Systems, OCR, Prompt Engineering, Embeddings, Workflow Automation, Document Ingestion
Data: PostgreSQL, SQL Server, Azure SQL, MySQL, ETL Pipelines, Data Analysis, Google Analytics
Deployment & Support: AWS, Azure, Docker, CI/CD, Sentry, Kibana, Log Analysis, PyTest, Git, Documentation, Customer Communication

Projects

Audit Evidence Matching Agent | *Python, FastAPI, LLM APIs, OCR, PostgreSQL, PyTest*

- Designed an **LLM workflow** that extracts fields from invoices and bank statements, matches them to ledger entries, and flags exceptions with source citations for reviewer sign-off and a full **audit trail**.

Customer Deployment Configuration Toolkit | *TypeScript, Node.js, REST APIs, PostgreSQL, Docker*

- Designed an **internal tool** that generates per-customer deployment configs for cloud and on-premise environments, runs pre-launch health checks against customer APIs, and produces **implementation runbooks** for onboarding.

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, SQL Server, Docker*

- Extended an AI-assisted **SQL** workspace supporting 10+ databases with schema exploration, AI-generated queries reaching 82% accuracy, and AI-assisted reporting dashboards for faster data analysis from one interface.